



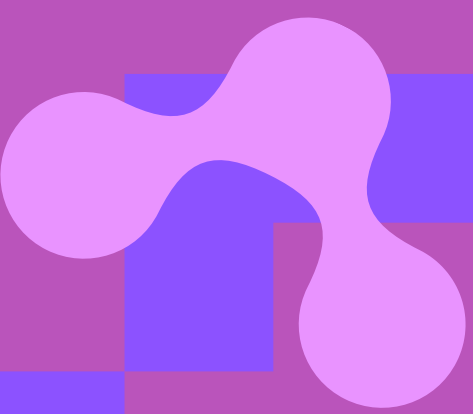
# ELECTRICAL SERVICE DESIGN EEE-316

**FABLIHA MAHAZABIN AHANA  
201 016 002**

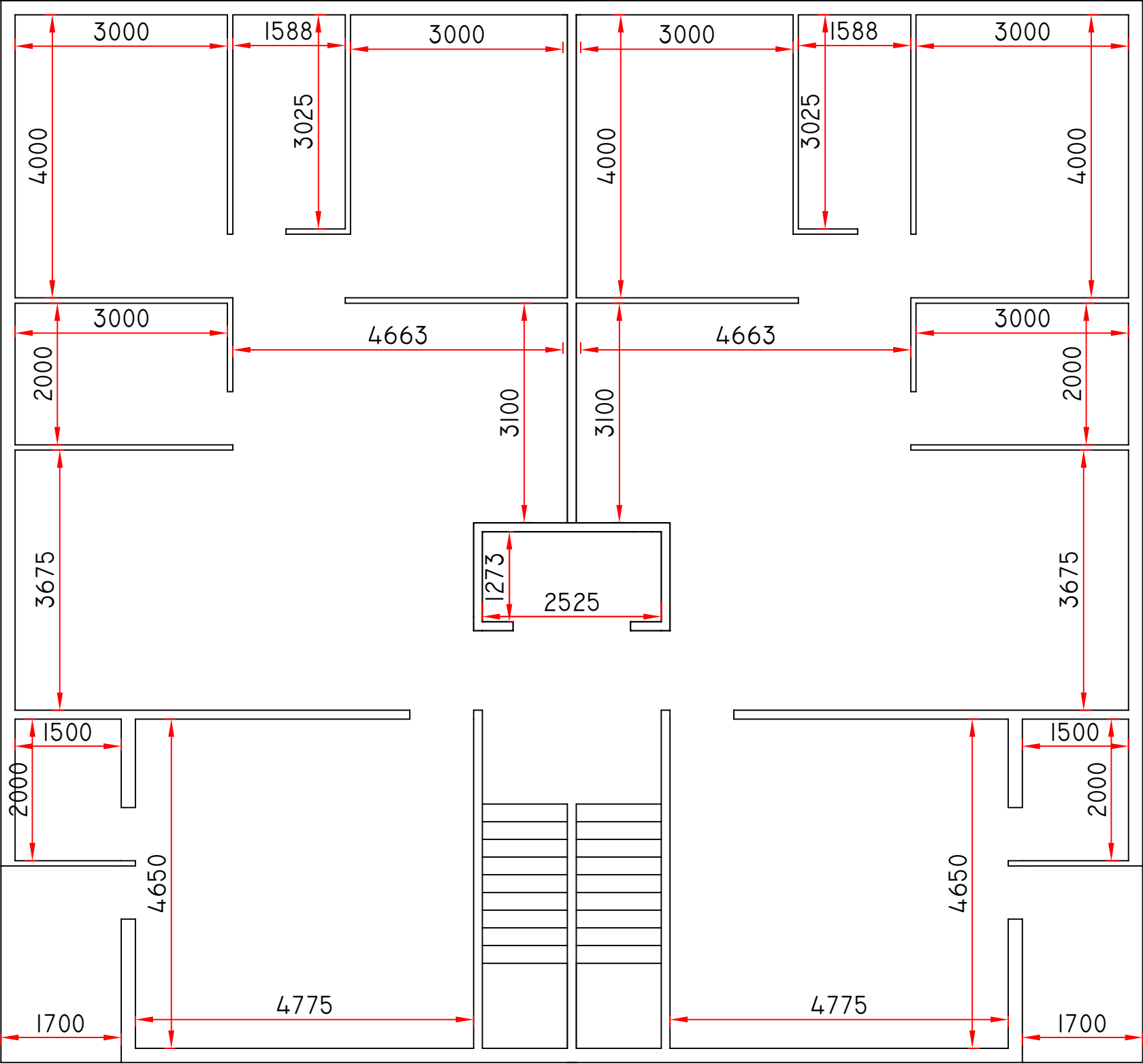
**&**

**SOWRAV CHOWDHURY  
183 016 008**

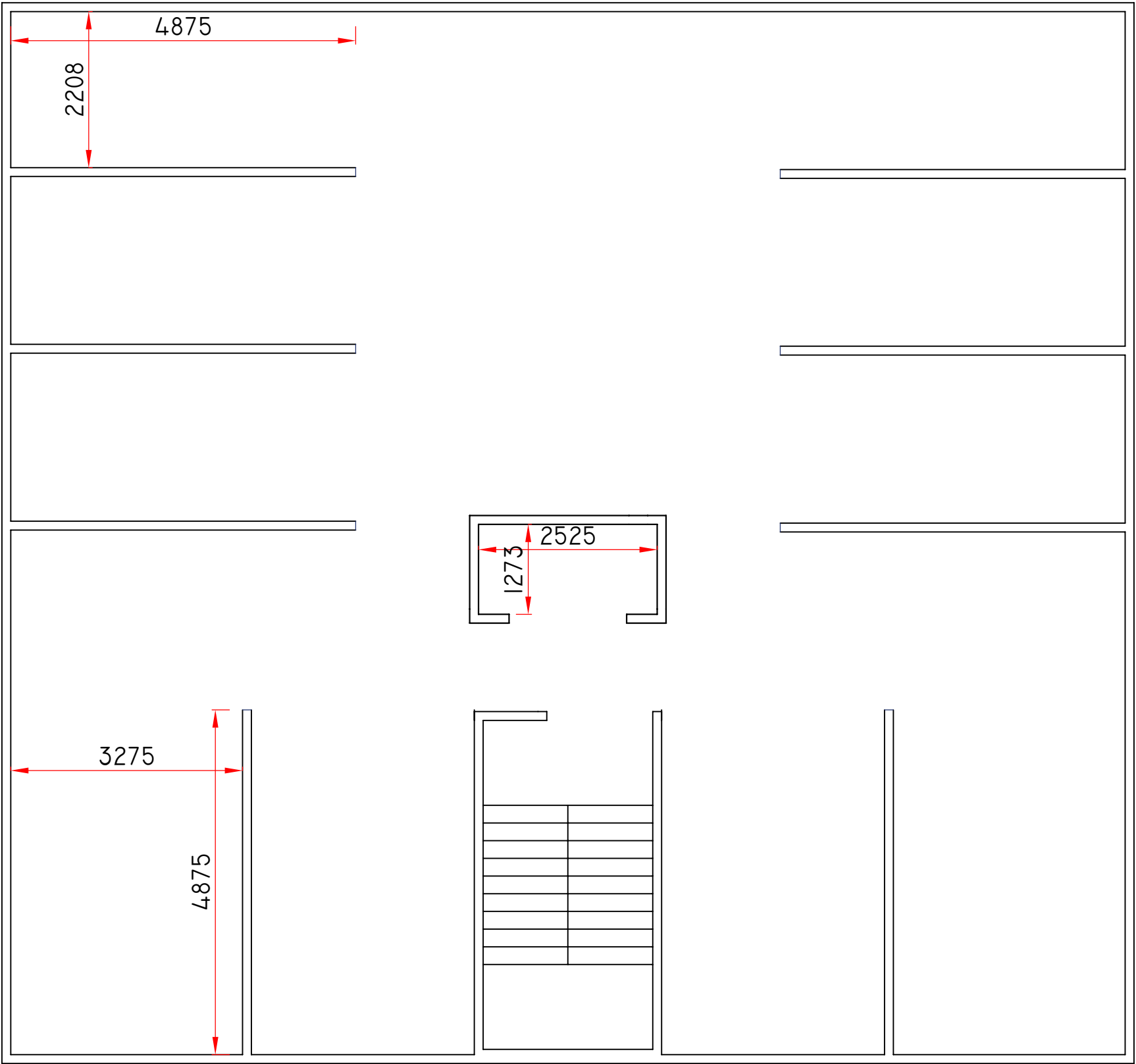
**DEPARTMENT OF ELECTRICAL AND ELECTRONICS  
ENGINEERING**



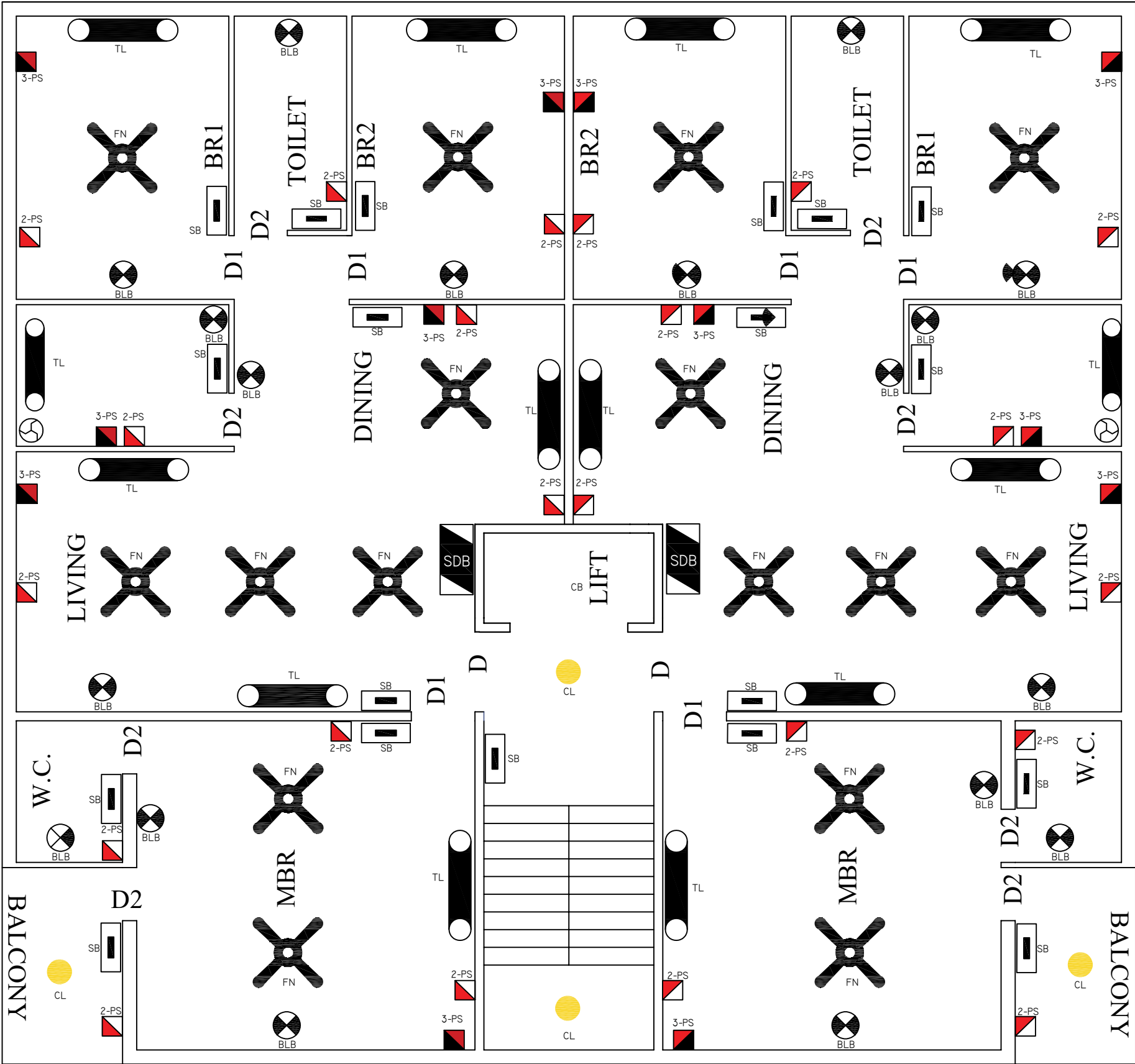
TYPICAL FLOOR PLAN



GROUND FLOOR PLAN

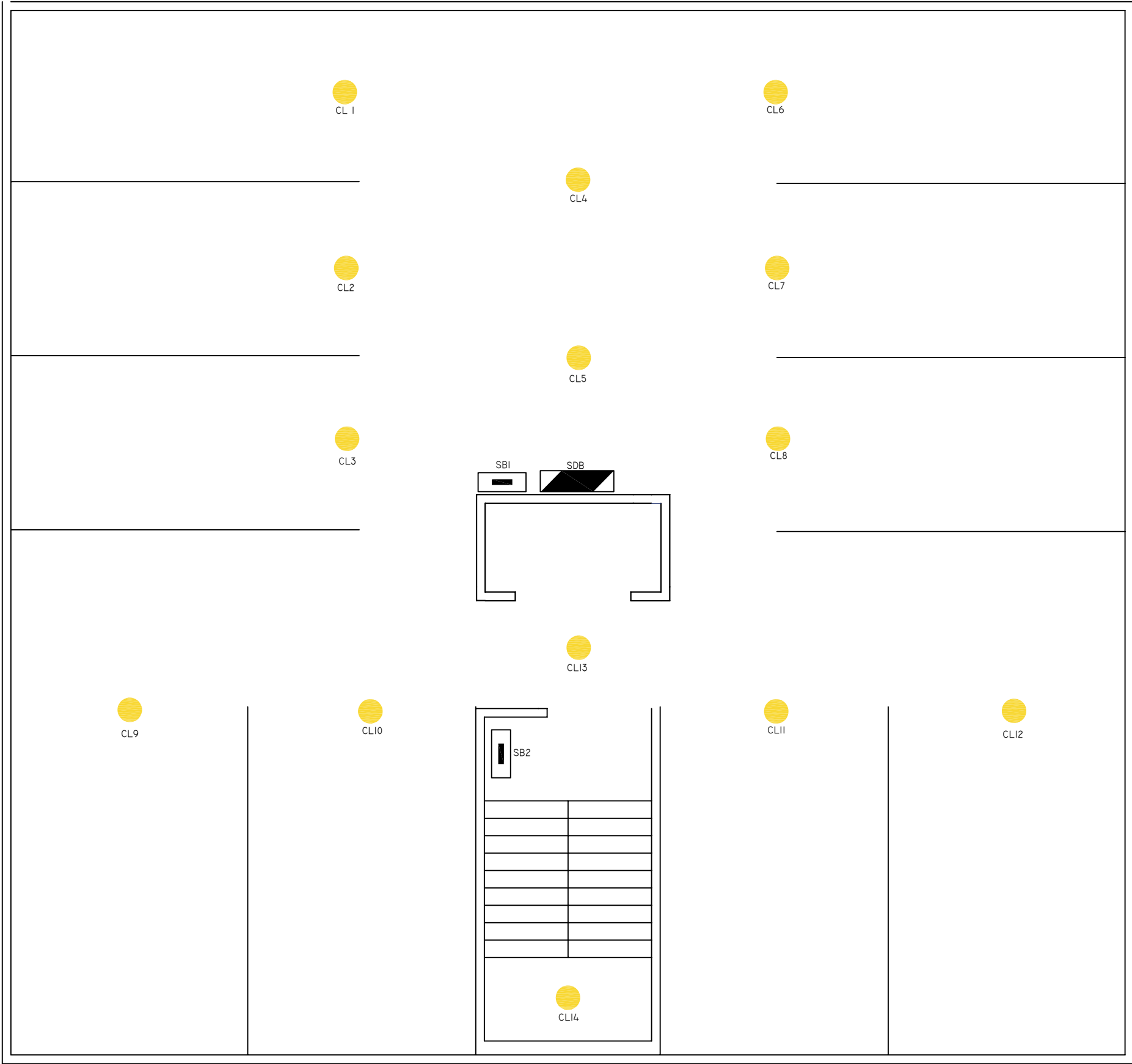


FITTINGS & FIXTURES  
TYPICAL FLOOR PLAN

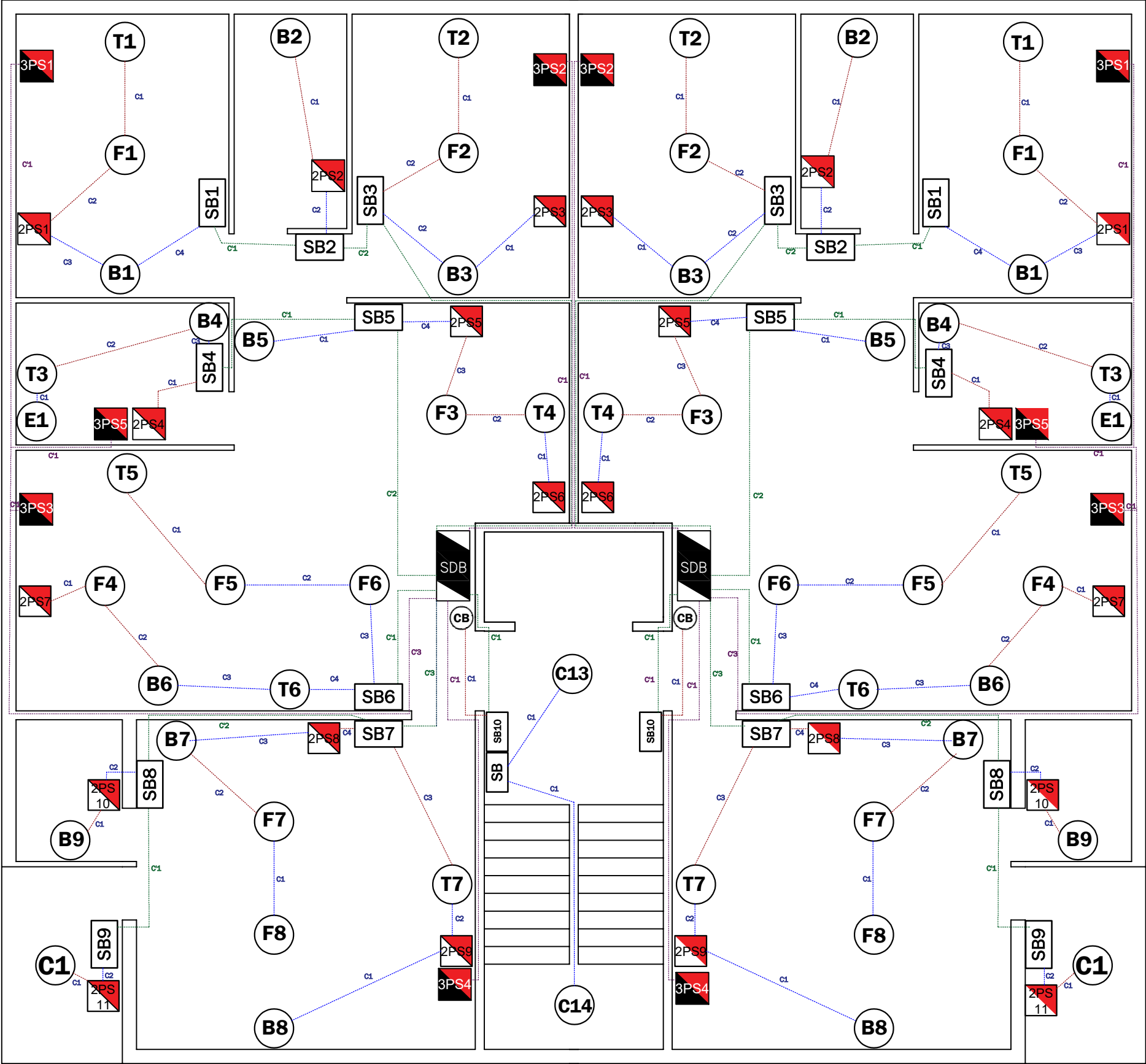


# FITTINGS & FIXTURES

## GROUND FLOOR PLAN



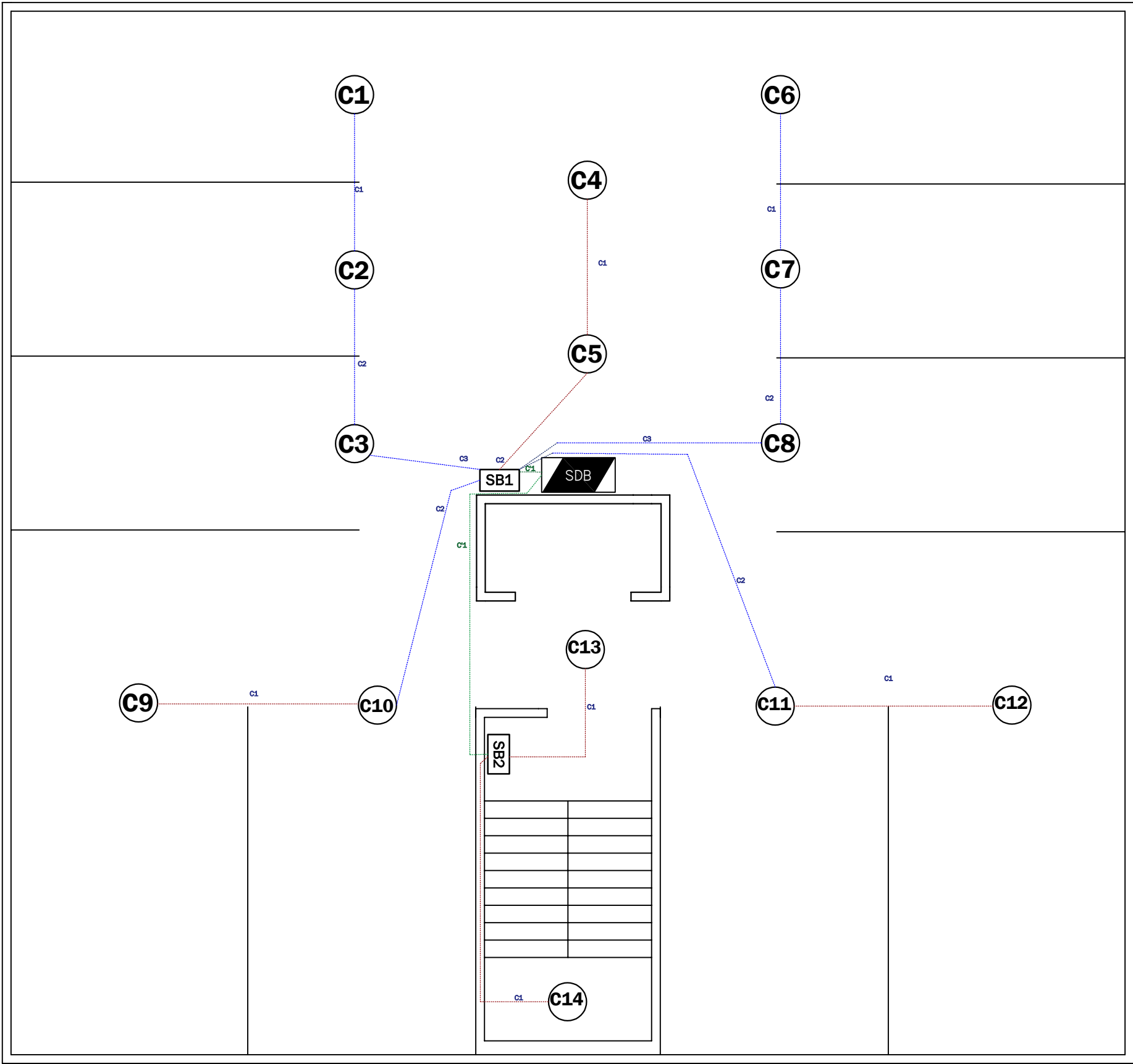
# CONDUIT LAYOUT TYPICAL FLOOR PLAN





# CONDUIT LAYOUT

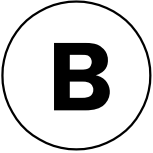
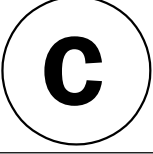
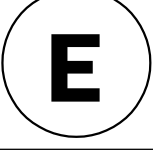
## GROUND FLOOR PLAN



LEGENDS AND INFORMATION

| SYMBOL                                                                              | NAME OF COMPONENTS                   | ABBREVIATIONS | POSITION   |
|-------------------------------------------------------------------------------------|--------------------------------------|---------------|------------|
|    | 40FT 20W WALL MOUNTED LED TUBE LIGHT | TL            | LINTEL     |
|    | 15W LED LIGHT BULB                   | BLB           | LINTEL     |
|    | 56" 75W SWEEP FAN                    | FN            | CEILING    |
|    | POWER SOCKET 2-PIN                   | 2-PS          | LOWER WALL |
|  | POWER SOCKET 3-PIN                   | 3-PS          | LINTEL     |
|  | SWITCH BOARD                         | SB            | MID WALL   |
|  | CALLING BELL                         | CB            | MID WALL   |
|  | EXHAUST FAN                          | EFN           | LINTEL     |
|  | 12W LED CEILING LIGHT                | CL            | CEILING    |
|  | SUB-DISTRIBUTION BOARD               | SDB           | LINTEL     |



| LEGENDS FOR CONDUIT                                                                 |                           |
|-------------------------------------------------------------------------------------|---------------------------|
|    | FAN                       |
|  | BULB                      |
|  | TUBE LIGHT                |
|  | CEILING LIGHT             |
|  | EXHAUST FAN               |
|  | SWITCH BOARD              |
|  | 2-PIN POWER<br>SOCKET     |
|  | 3-PIN POWER<br>SOCKET     |
|  | SUB-DISTRIBUTION<br>BOARD |

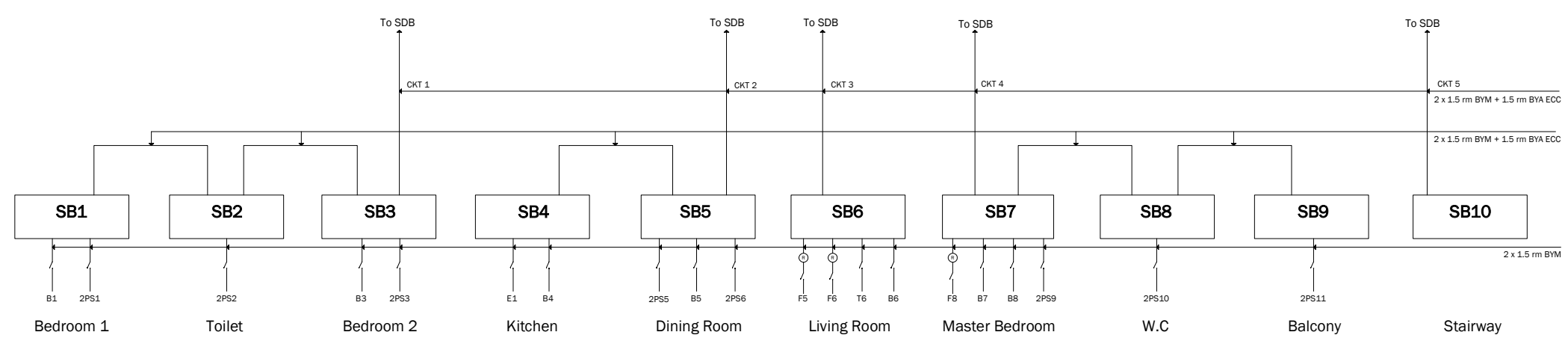
**Legends for Conduits**

- C'3 = 6 x 1.5 rm BYM + 1.5 rm BYA ECC
- C'2 = 4 x 1.5 rm BYM + 1.5 rm BYA ECC
- C'1 = 2 x 1.5 rm BYM + 1.5 rm BYA ECC
- C4 = 8 x 1.5 rm BYM
- C3 = 6 x 1.5 rm BYM
- C2 = 4 x 1.5 rm BYM
- C1 = 2 x 1.5 rm BYM

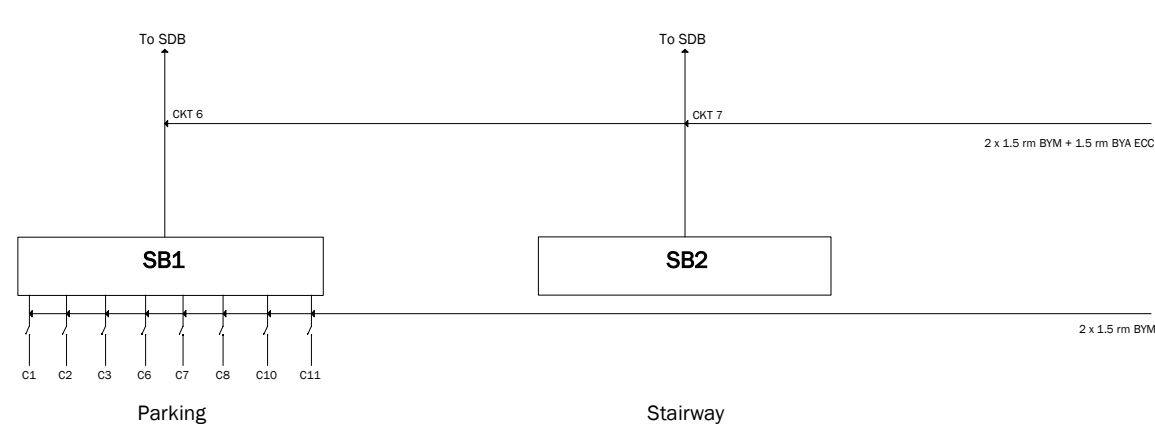
**Conduit Size**

- 1"
- 0.75"
- 0.75"
- 1"
- 0.75"
- 0.75"
- 0.75"

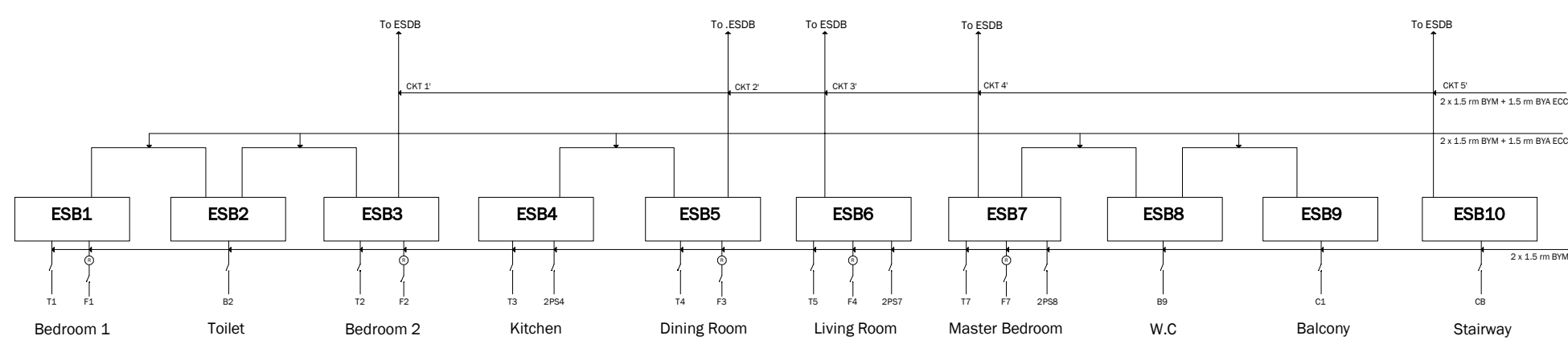
SWITCH BOARD CONNECTION DIAGRAM OF  
TYPICAL FLOOR PLAN



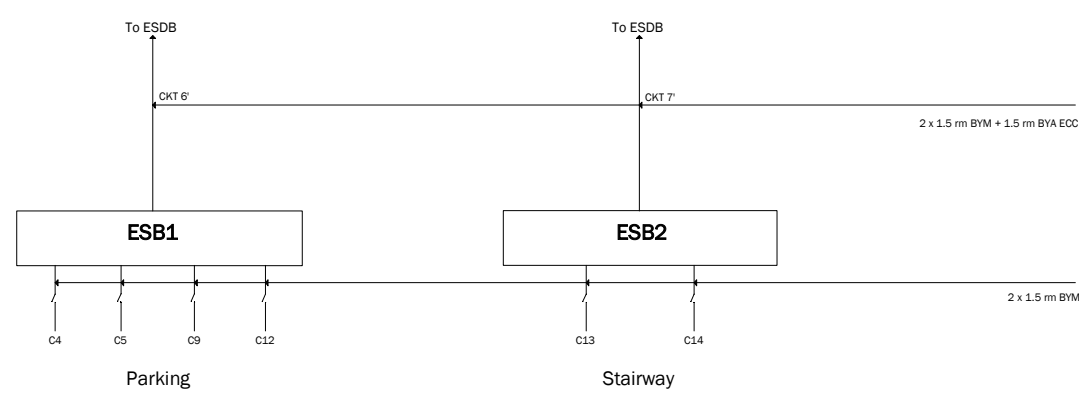
SWITCH BOARD CONNECTION DIAGRAM OF  
GROUND FLOOR PLAN



EMERGENCY SWITCH BOARD  
CONNECTION DIAGRAM OF  
TYPICAL FLOOR PLAN



EMERGENCY SWITCH BOARD  
CONNECTION DIAGRAM OF  
GROUND FLOOR PLAN



## Calculations for Lights (TL, BLB & CL) and Fans (FN & EFN)

Formula for Lights,  $E = \frac{n \times N \times F \times UF \times LLF}{A}$ ; (A in m<sup>2</sup>)

One 56" diameter fan is needed every 100 sqft.

Number of Fans =  $\frac{A}{100}$ ; (A in sqft)

### I. Bedroom 1 & 2

Area = 13.12 × 9.84 sqft = 12 m<sup>2</sup>;

Illuminance,  $E = 150$  Lumen/m<sup>2</sup>;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 2400 Lumen (20W 4ft LED Tube light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 1.07$

So, 1 Light Bulbs and 1 Tube Light are needed.

Number of Fans = 1.29

So, 1 fan is needed.

### II. Master Bedroom

Area = 15.67 × 15.26 sqft = 22.204 m<sup>2</sup>;

Illuminance,  $E = 150$  Lumen/m<sup>2</sup>;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 2400 Lumen (20W 4ft LED Tube light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 1.98$

So, 2 Light Bulbs and 1 Tube Light are needed.

Number of Fans = 2.38

So, 2 fans are needed.

### III. Living room

Area = 24.20 × 12.04 sqft = 27.103 m<sup>2</sup>;

Illuminance,  $E = 200$  Lumen/m<sup>2</sup>;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 2400 Lumen (20W 4ft LED Tube light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 3.23$

So, 1 Light Bulb and 2 Tube Lights are needed.

Number of Fans = 2.91

So, 3 fans are needed.

#### **IV. Dining room**

Area =  $15.30 \times 10.17$  sqft =  $14.455 \text{ m}^2$ ;

Illuminance,  $E = 200 \text{ Lumen/m}^2$ ;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 2400 Lumen (20W 4ft LED Tube light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 1.72$

So, 1 Light Bulbs and 1 Tube Light are needed.

Number of Fans = 1.55

So, 1 fan is needed.

#### **V. Kitchen**

Area =  $6.56 \times 1.64$  sqft =  $6 \text{ m}^2$ ;

Illuminance,  $E = 300 \text{ Lumen/m}^2$ ;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 2400 Lumen (20W 4ft LED Tube light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 1.07$

So, 1 Light Bulbs and 1 Tube Light are needed.

1 Exhaust Fan is needed.

#### **VI. Toilet**

Area =  $5.21 \times 9.92$  sqft =  $4.80 \text{ m}^2$ ;

Illuminance,  $E = 100 \text{ Lumen/m}^2$ ;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 1500 Lumen (15W LED Light Bulb);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 0.45$

So, 1 Light Bulb is needed.

#### **VII. W.C**

Area =  $6.56 \times 1.64$  sqft =  $1 \text{ m}^2$ ;

Illuminance,  $E = 100 \text{ Lumen/m}^2$ ;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 1500 Lumen (15W LED Light Bulb);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 0.10$

So, 1 Light Bulb is needed.



### **VIII. Balcony**

Area =  $5.58 \times 8.32$  sqft =  $4.31 \text{ m}^2$ ;

Illuminance,  $E = 100$  Lumen/ $\text{m}^2$ ;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 1200 Lumen (12W LED ceiling light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 0.51$

So, 1 Ceiling Light is needed.

### **IX. Stairway**

Area =  $7.30 \times 16.40$  sqft =  $11.13 \text{ m}^2$ ;

Illuminance,  $E = 100$  Lumen/ $\text{m}^2$ ;

Light Loss Factor and Utilization Factor,  $LLF \times UF = 0.7$ ;

Number of lights per luminaire,  $n = 1$ ;

Flux = 1200 Lumen (12W LED ceiling light);

Number of Lights,  $N = ?$

Calculating from the above formula,  $N = 1.33$

So, 2 Ceiling Light is needed.

## Calculation for Conduits

Formula for Ampere Rating,  $I = \frac{P}{V \times pf}$  (A)

$Pf = 0.7$  and Voltage = 220 V are considered on an average.

LED Bulb = 15 W

Tube Light = 20 W

Ceiling Fan = 75 W

Switchboard Socket (max) = 100 W

Ceiling Light = 12 W

Exhaust Fan = 35 W

Calling Bell = 15 W

All Internal wires are below 5 A rating so 2 x 1.5 rm BYM is used in all internal wiring.

## To Sub Distribution Board (SDB) (Typical Floor Plan)

### CKT1 Rating:

$$I = \frac{15 \times 2 + 100 \times 3}{220 \times 0.7} = 2.14A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

### CKT2 Rating:

$$I = \frac{35 + 15 \times 2 + 100 \times 2}{220 \times 0.7} = 1.72A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

### CKT3 Rating:

$$I = \frac{20 + 75 \times 2 + 15}{220 \times 0.7} = 1.20A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

### CKT4 Rating:

$$I = \frac{75 + 100 \times 3 + 15 \times 2}{220 \times 0.7} = 2.63A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

### CKT5 Rating:

$$I = \frac{0}{220 \times 0.7} = 0A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

## To Sub Distribution Board (SDB) (Ground Floor Plan)

### CKT6 Rating:

$$I = \frac{12 \times 8}{220 \times 0.7} = 0.62A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

### CKT7 Rating:

$$I = \frac{0}{220 \times 0.7} = 0A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.



### **To Emergency Sub Distribution Board (SDB) (Typical Floor Plan)**

#### **CKT1' Rating:**

$$I = \frac{20 \times 2 + 75 \times 2 + 15}{220 \times 0.7} = 1.33A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

#### **CKT2' Rating:**

$$I = \frac{20 \times 2 + 100 + 75}{220 \times 0.7} = 1.40A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

#### **CKT3' Rating:**

$$I = \frac{20 + 75 + 100}{220 \times 0.7} = 1.27A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

#### **CKT4' Rating:**

$$I = \frac{20 + 75 + 100 + 15 + 12}{220 \times 0.7} = 1.44A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

#### **CKT5' Rating:**

$$I = \frac{15}{220 \times 0.7} = 0.09A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

### **To Emergency Sub Distribution Board (SDB) (Ground Floor Plan)**

#### **CKT6' Rating:**

$$I = \frac{12 \times 4}{220 \times 0.7} = 0.31A$$

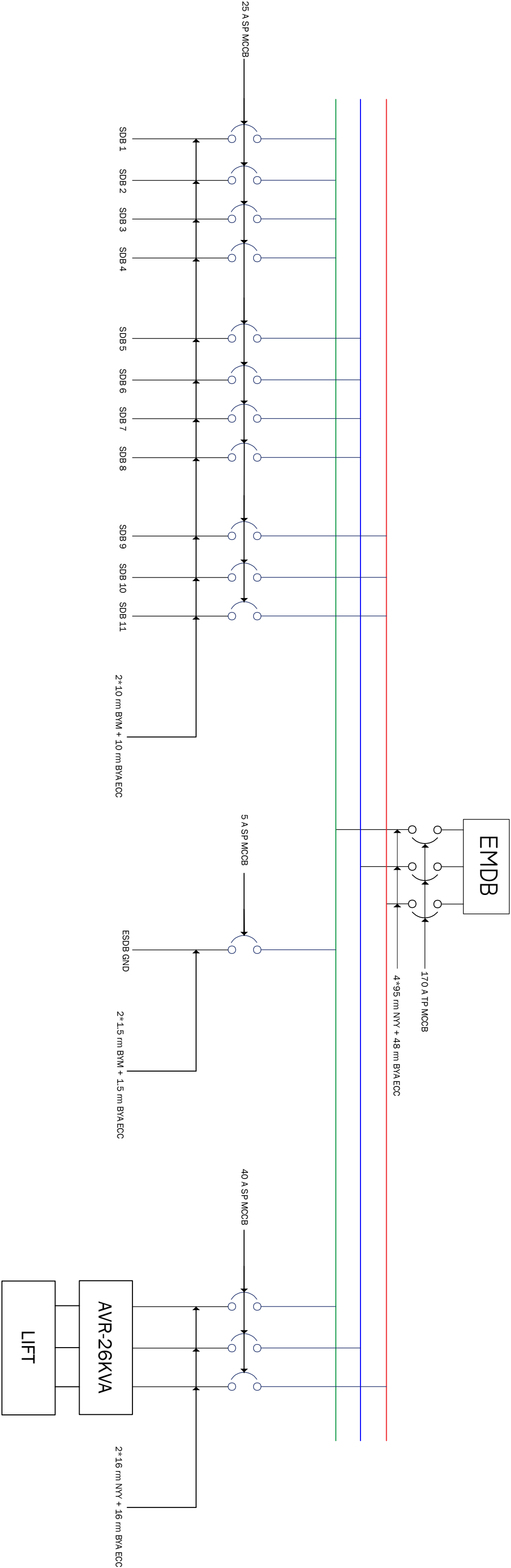
So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

#### **CKT7' Rating:**

$$I = \frac{12 \times 2}{220 \times 0.7} = 0.15A$$

So, 2 x 1.5 rm BYM + 1.5 BYA ECC are used.

EMERGENCY MAIN DISTRIBUTION BOARD DIAGRAM



MAIN DISTRIBUTION BOARD DIAGRAM

